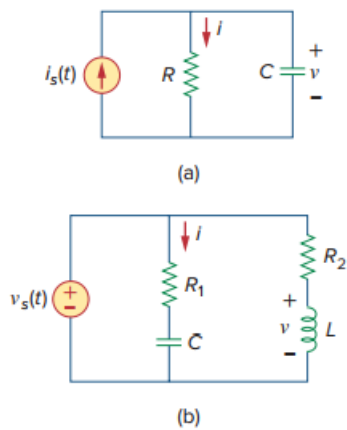


Problem

Using Fig. 9.45, design a problem to help other students better understand admittance.

Figure 9.45:



Step-by-step solution

Step 1 of 5

(a) Given circuit can be represented as,

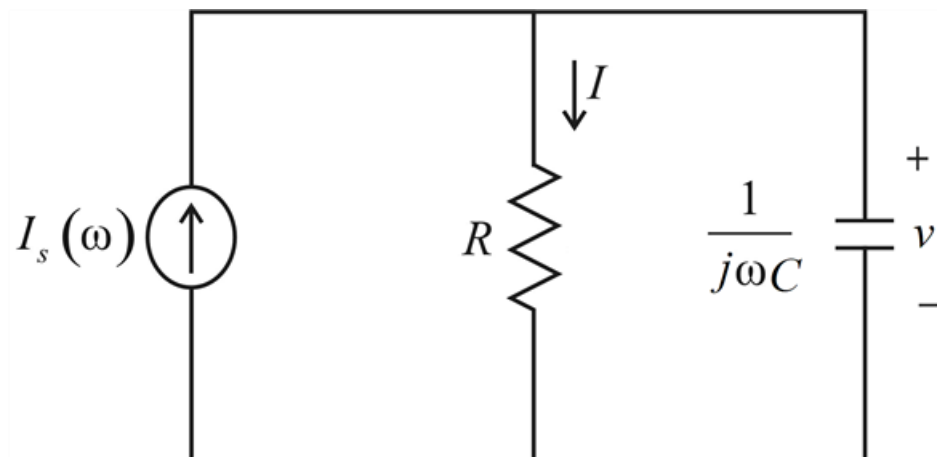


Figure (1)

Step 2 of 5

From the above circuit,

$$v = \frac{I_s(\omega)}{Y_{equ}}$$

Where,

$$Y_{equ} = Y_R + Y_C$$

$$Y_{equ} = \frac{1}{R} + \frac{1}{\frac{1}{j\omega C}}$$

$$Y_{equ} = \frac{1}{R} + j\omega C$$

Hence,

$$v = \frac{I_s(\omega)}{\frac{1}{R} + j\omega C}$$

$$\therefore v = \frac{I_s(\omega) \times R}{1 + j\omega RC} \text{ V}$$

Step 3 of 5

From Figure (1)

$$I = v \times Y_{equ}$$

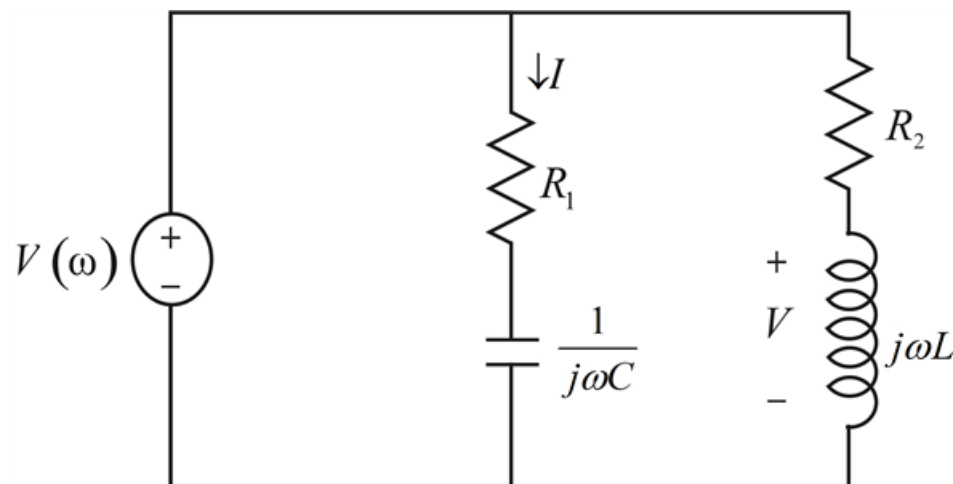
$$I = v \times Y_R$$

$$I = \left(\frac{I_s(\omega) \times R}{j\omega RC + 1} \right) \times \frac{1}{R}$$

$$\therefore I = \frac{I_s(\omega)}{1 + j\omega RC} \text{ A}$$

Step 4 of 5

(b) Given circuit can be represented as,



From the above circuit,

$$i(\omega) = V(\omega) Y_{RC}$$

$$Y_{RC} = \frac{1}{R_1 + \frac{1}{j\omega C}}$$

$$I = \frac{V(\omega)}{R_1 + \frac{1}{j\omega C}}$$

$$\therefore I = \frac{V(\omega)(j\omega L)}{1 + j\omega CR_1} \text{ A}$$

By using voltage division principle to above circuit, we get

$$V = V(\omega) \times \left[\frac{j\omega L}{R_2 + j\omega L} \right] \text{ V}$$